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An integrated machine learning and MARCOS method for supplier evaluation and selection

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ABSTRACT

Evaluating and selecting suitable suppliers involves several criteria, making it a complex and difficult problem for decision-makers. Several approaches have been developed to help achieve the best possible results for selecting the most suitable suppliers. Traditionally, multi-criteria decision-making methods, such as the recently introduced MARCOS (Measurement of Alternatives and Ranking according to COmpromise Solution) method, have been used for supplier evaluation and selection. A common issue to be addressed in such methods is determining appropriate weights for the evaluation criteria. We propose an integrated approach that combines machine learning with the MARCOS method for this evaluation process. The approach uses feature importance methods based on interpretable tree-based machine learning to calculate weights for supplier selection criteria. These weights are then used as part of the MARCOS method to rank individual suppliers and select the supplier with the highest rank. The proposed approach is validated through a real-world case study in the oil and gas sector and is compared with state-of-the-art approaches in multi-criteria decision-making and machine learning-based classification. Results show that the proposed model selects suppliers in line with human evaluators and performs similarly to other multi-criteria decision-making approaches. The proposed approach can be useful for designing effective and efficient supplier evaluation systems.

1. Introduction

Supplier selection is an important step in the procurement process, through which purchasing managers choose the best suppliers for the services or raw materials they wish to purchase. Companies typically seek out the best suppliers to improve performance and maintain existing relationships over the long term. In addition, in many industries, the value of raw materials represents the largest portion of manufacturing costs [1], making supplier selection a crucial factor in ensuring financial stability of a company. Additionally, choosing the right suppliers reduces purchasing risks while increasing consumer satisfaction and developing positive relationships between buyers and suppliers [2]. On the other hand, choosing the wrong suppliers can cause economic problems and affect a company's performance.

The supplier selection process is a strategic supply chain purchasing management activity and typically consists of the following six steps [3]:

- Understand business priorities and strategies,
- Determine the criteria used in the selection,
- Calculate the importance of criteria,
- Identify potential suppliers,
- Rank the suppliers according to the specified criteria,
- Select one or more suppliers based on the calculated ranking.

Business priorities and strategies are the main factors that determine which criteria are to be used for comparison and how these criteria will be measured. Supplier ranking is generally the responsibility of purchasing managers, who may consider different classifications of the evaluation criteria. Therefore, the definition and weighting of supplier selection criteria are two of the most important steps of the supplier selection process. After determining potential suppliers and selection criteria, the following phase is to select the most qualified potential suppliers.

The rising complexity of global supply networks is reflected in important decision-making processes, including the selection of suppliers [4]. While the complexity of determining selection criteria based on business priorities can scale well since the number of stakeholders involved in this step is usually limited, the process of calculating criteria importance and ranking suppliers is significantly affected as the number of suppliers and criteria increases. The combinatorial nature of these problems has traditionally led researchers and practitioners to utilise multi-criteria decision making (MCDM) approaches in order to develop efficient solutions. However, while MCDM approaches afford

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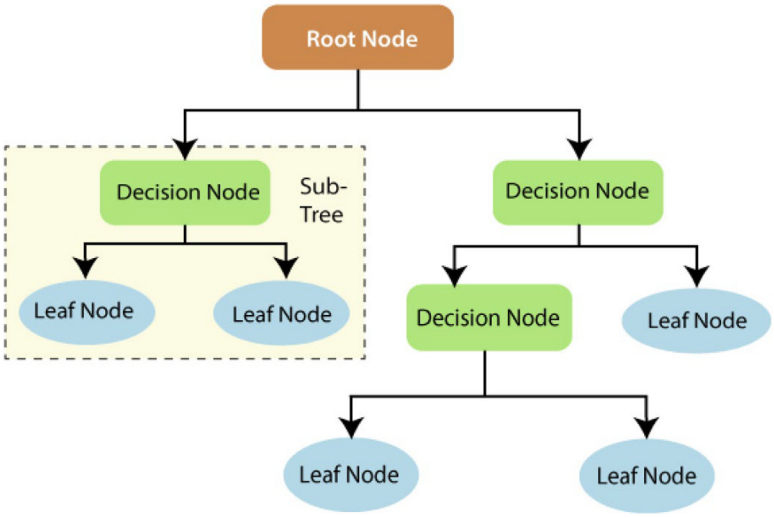

Fig. 1. Standard decision tree structure [7].

more scalability than a fully manual supplier selection process, they are still limited in large-scale scenarios with many different but potentially correlated and competing criteria and large amounts of suppliers. With supply chains and supply networks [5] becoming increasingly globalised, approaches that rely on large amounts of data, such as machine learning (ML), have also been explored to address the supplier selection problem [6].

MCDM and ML have different requirements and at the same time offer different ranges of benefits. As such, combinations between them can provide opportunities to address more requirements and have access to more benefits, while also being able to mitigate drawbacks faced individually by each type of approach. This work aims to present such an integrated approach, combining machine learning approaches that are used to calculate selection criteria weights, which are then used as part of the recently proposed Measurement of Alternatives and Ranking according to COmpromise Solution (MARCOS) method [8] to rank suppliers. The proposed approach is validated through a real-world case study with a company in the oil and gas industry. The contributions of this study can be summarised as follows:

- A novel integrated approach for supplier selection that combines machine learning classification with the MARCOS method,
- A demonstration the proposed approach through a case study that validates its outcomes against decisions made by procurement managers,
- A comparative evaluation of different classification algorithms to determine suitability in the context of the proposed approach,
- Sensitivity analysis confirms that the proposed integrated approach is capable of achieving results in line with other MCDM approaches used in literature for supplier selection.

The remainder of this paper is structured as follows. Section 2 summarises relevant background knowledge and provides a focused overview of related literature in integrated approaches for supplier selection. Section 3 is devoted to discussing the proposed integrated approach. In this section, the four methods used in the paper for classification are discussed, and the MARCOS method that is applied in ranking suppliers is explained. Section 4 presents a case study that was conducted with an oil and gas company in Libya to validate the proposed approach. Finally, Section 5 concludes and points out several directions for future research.

2. Background and related research

This section first provides an overview of the relevant research background in terms of MCDM and ML, before then summarising supplier selection research that is directly related to this study.

2.1. Multi-criteria decision making

In making decisions that involve prioritising, ranking, or selecting between preferences, researchers utilise techniques that are collectively known as multi-criteria decision making approaches. Their objective is to assist decision-makers in selecting preferred options or directly select single alternative that meets their needs in the most optimal manner. MCDM approaches incorporate preferences over a range of quantitative and qualitative criteria which, in most cases, are competing and are not easily reconciled, complicating the process of reaching an agreed decision. Different disciplines inform the development of MCDM techniques, including information systems, economics, computer science and behavioural decision theory.

In literature, a diverse array of MCDM techniques has been proposed. These include the Best Worst Method (BWM) [9] and Decision making trial and evaluation laboratory (DEMATEL) [10], Logarithmic Percentage Change driven Compromise Solution based Appraisal (LOPCCSA) [11] and the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) [12].

Each of the approaches contributes differently to the same issue of being able to make decisions in the face of competing priorities, and each has its own unique privileges [13]. Despite the many studies that have been conducted and solutions that have been implemented, MCDM is still quite actively researched and applied in a wide variety of problems, from renewable energy strategies [14] to supplier selection [15].

2.2. ML classification

There is a broad range of ML algorithms that can be used for classification. In this section, we only present an overview of four of these that are considered in the proposed approach: decision trees, random forests, extremely randomised trees and gradient boosting through CatBoost (please refer to Section 3.2 for the rationale behind selecting these). We also briefly introduce feature importance, which is used as the main mechanism of calculating criteria weights in the proposed approach.

Decision trees are extremely versatile ML algorithms used to do classification and regression problems, as well as multioutput tasks. Their ability to accommodate a wide range of datasets, as well the inherent understandability of a tree structure makes them one of the most commonly used approaches [16]. As shown in Fig. 1, a typical tree involves a recursive structure of decision nodes, beginning with a root node, moving through zero or more internal decision nodes and

Fig. 2. Typical Random Forest classification through majority voting [20].

ending with one or more leaf nodes. At each decision node, the decision to split is based on pre-specified criteria and depends on whether variables are quantitative or qualitative. For qualitative (or categorical variables), splitting aims to maximise the homogeneity of the two subtrees, through metrics such as Information Gain or Gini Impurity. When leaf nodes are reached, the class that represents the majority between the contained samples is chosen.

Random Forests are a generalisation of decision trees and, in essence, are a collection of multiple tree instances. Despite its simplicity, Random Forest classification is very effective [16]. Random Forests depend on training several Decision

Tree algorithms, each of which is trained on a unique random subset of the training set, and then the results are combined. Commonly, results combination is conducted through a majority voting process. Fig. 2 illustrates this process, where each decision tree results in a particular class, and the final class chosen is the one the majority have resulted in, in this example class C. Extremely randomised trees [17] (also known as Extra Trees) expand on the main idea of Random Forests to develop a more computationally efficient ensemble learning approach. Extra Trees builds an ensemble of unpruned classification/regression trees with a traditional top-down procedure. In addition, a random subset of features is employed to train each base estimator, which is identical to the Random Forest algorithm. However, rather than selecting the most discriminating split in each node, the algorithm randomly chooses the best feature and matching value for splitting the node [18]. This increases performance, as splitting logic is far simpler, however this also makes the algorithm less robust to noise. Effectiveness and accuracy of different classification approaches varies and extensive research has been conducted in optimising them [19].

Finally, another ensemble learning approach typically associated with trees is gradient boosting [21]. As an ensemble method, it is similar to Random Forests and Extra Trees in terms of creating several models (decision trees in the particular case of gradient tree boosting) and combining their results. In contrast to those, however, the creation of these models is iterative, with each iteration aiming to minimise the loss function. In the case of datasets that include categorical features, the CatBoost framework has been proposed [22]. CatBoost aims to address issues faced by other gradient boosting algorithms by implementing ordered boosting, where training data is ordered according to a random permutation and n models are trained using only a subset of the training set, with model M_i using only the first i samples.

2.3. Related work

The extensive amount of research that has been carried out based on the applications of various MCDM methods and their integration with other methods in various manufacturing domains illustrates the significance of the supplier selection process. This section presents selected related work on the supplier selection and evaluation problem that has been addressed using MCDM approaches (primarily ones integrating multiple MCDM approaches), ML algorithms, or their combinations. A critical evaluation is presented, focusing primarily on recently published research and aiming to motivate the particular research gap that the methodology proposed in Section 3 intends to address. For a more systematic analysis of supplier selection literature, the reader is referred to recently published literature review papers, such as those by Echefaj et al. [23] and Song, Xu, & Xu [24]

A relatively early example of integration of different MCDM approaches to create a decision support system for supplier selection is the work of Orji & Wei [25]. The Fuzzy Decision-Making Trial and Evaluation Laboratory (FDEMATEL) and TOPSIS were combined to apply a sustainable selection of suppliers to a company in China that was responsible for the production of gears. The proposed model is able to effectively integrate aspects of sustainability into the supplier selection process by accurately capturing all relevant subjective and objective criteria present in the relevant environment.

Kaushik, Kumar, Gupta, & Dixit [26] proposed a hybrid supplier selection methodology that combines the Best Worst Method (BWM) and the VIKOR approach. The methodology was validated through a case study in the online fashion retail domain. Out of 38 selection criteria those that were selected as most important were Just-in-time (JIT) capabilities, order flexibility, return policies, technical soundness, and billing flexibility. The study's findings revealed managerial and theoretical implications for the chosen domain.

A number of integrated MCDM approaches have been proposed that include the MARCOS method, which is also included in the methodology that is the focus of this work. Badi & Pamucar [27], presented a hybrid Grey theory-MARCOS strategy that was applied to a particular company in the steel manufacturing sector. COmbinative Distance-based ASsessment (CODAS), TOPSIS, and VIKOR were used to compare the model's effectiveness. The model was able to identify the best supplier among six alternatives. Chakraborty, Chattopadhyay, & Chakraborty [28] used D numbers with MARCOS to choose raw material suppliers in India's iron and steel manufacturing. Three decision-makers with diverse knowledge and experience evaluated the relative performance of five competing suppliers with seven criteria using D-numbers. The MARCOS approach was then used to rank suppliers.

Moving to ML algorithms, as with several other application areas, research that applies ML to solve supplier selection problems has rapidly increased in recent years. Relatively recent examples include